hemicellulose, cellulose and callose together with proteins. Exine is thick and is made up of cellulose, sporopollenin and pollenkitt. The exine is not uniform and is thin at certain areas. When these thin areas are small and round it is called germ pores or when elongated it is called furrows. It is associated with germination of pollen grains. The sporopollenin is generally absent in germ pores.The surface of the exine is either smooth or sculptured in various patterns (rod like, grooved, warty, punctuate etc.) The sculpturing pattern is used in the plant identification and classification.

Shape of a pollen grain varies from species to species. It may be globose, ellipsoid, fusiform, lobed, angular or crescent shaped. The size of the pollen varies from 10 micrometers in *Myosotis* to 200 micrometers in members of the family Cucurbitaceae and Nyctaginaceae

DO YOU KNOW?

Palynology is the study of pollen grains.It helps to identify the distribution of coal and to locate oil fields. Pollen grains reflect the vegetation of an area.

Liquid nitrogen (-196^{0}C) is used to preserve pollen in viable condition for prolonged duration. This technique is called **cryopreservation** and is used to store pollen grains (pollen banks) of economically important crops for breeding programmes..

The wall material sporopollenin is contributed by both pollen cytoplasm and tapetum. It is derived from carotenoids. It is resistant to physical and biological decomposition. It helps to withstand high temperature and is resistant to strong acid, alkali and enzyme action. Hence, it preserves the pollen for long periods in fossil deposits, and it also protects pollen during its journey from anther to stigma.

Pollenkitt is contributed by the tapetum and coloured yellow or orange and is chiefly made of carotenoids or flavonoids. It is an oily layer forming a thick viscous coating over pollen surface. It attracts insects and protects damage from UV radiation.

DO YOU KNOW?

Bee pollen is a natural substance and contains high protein, carbohydrate, trace amount of minerals and vitamins. Therefore, it is used as dietary supplement and is sold as pollen tablets and syrups. Further , it increases the performance of athletes, race horses and also heals the wounds caused by burns . The study of honey pollen is called Mellitopalynology.



Development of Male gametophyte:

The microspore is the first cell of the male gametophyte and is haploid. The development of male gametophyte takes place while they are still in the microsporangium. The nucleus of the microspore divides mitotically to form a **vegetative** and a **generative** nucleus. A wall is laid around the generative nucleus resulting in the formation of two unequal cells, a large irregular nucleus bearing with abundant food reserve called vegetative cell and a small generative cell. Generally at this 2 celled stage, the pollens are liberated from the anther. In some plants the generative cell again undergoes a division to form two male gametes. The pollen is liberated at this 3 celled stage. In 60% of the angiosperms pollen is liberated in 2 celled stage. Further, the growth of the male gametophyte occurs only if the pollen reaches the right stigma. The pollen on reaching the stigma absorbs moisture and swells. The intine grows as pollen tube through the germ pore. In case the pollen is liberated at 2 celled stage the generative cell divides in the pollen into 2 male cells (sperms) after reaching the stigma or in the pollen tube before reaching the embryo sac. The stages in the development of male gametophyte is given in Figure 1.6.

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